Computer Organization and Architecture
(Code: CS256)

Lecture-7: CPU Organization II

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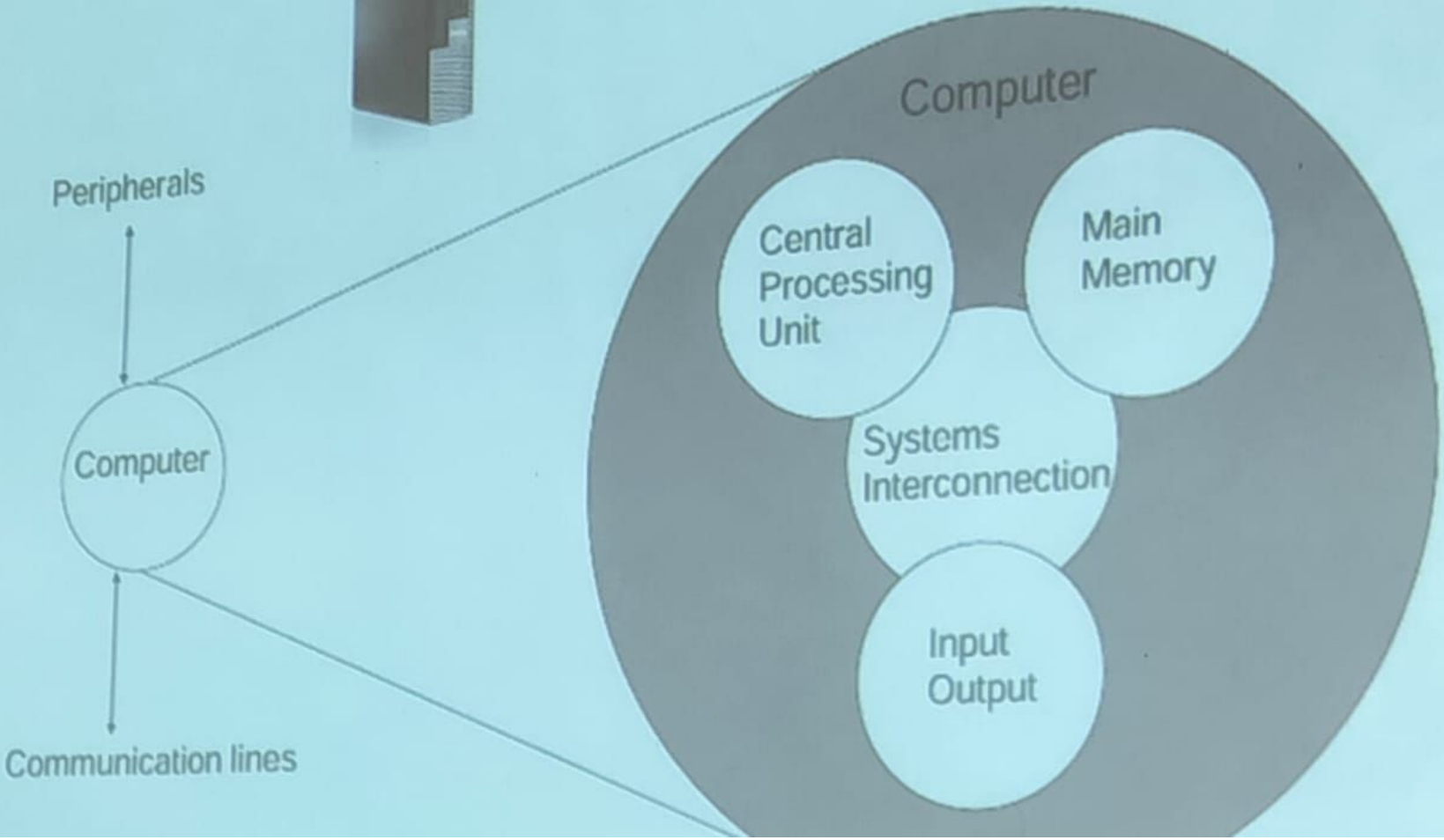
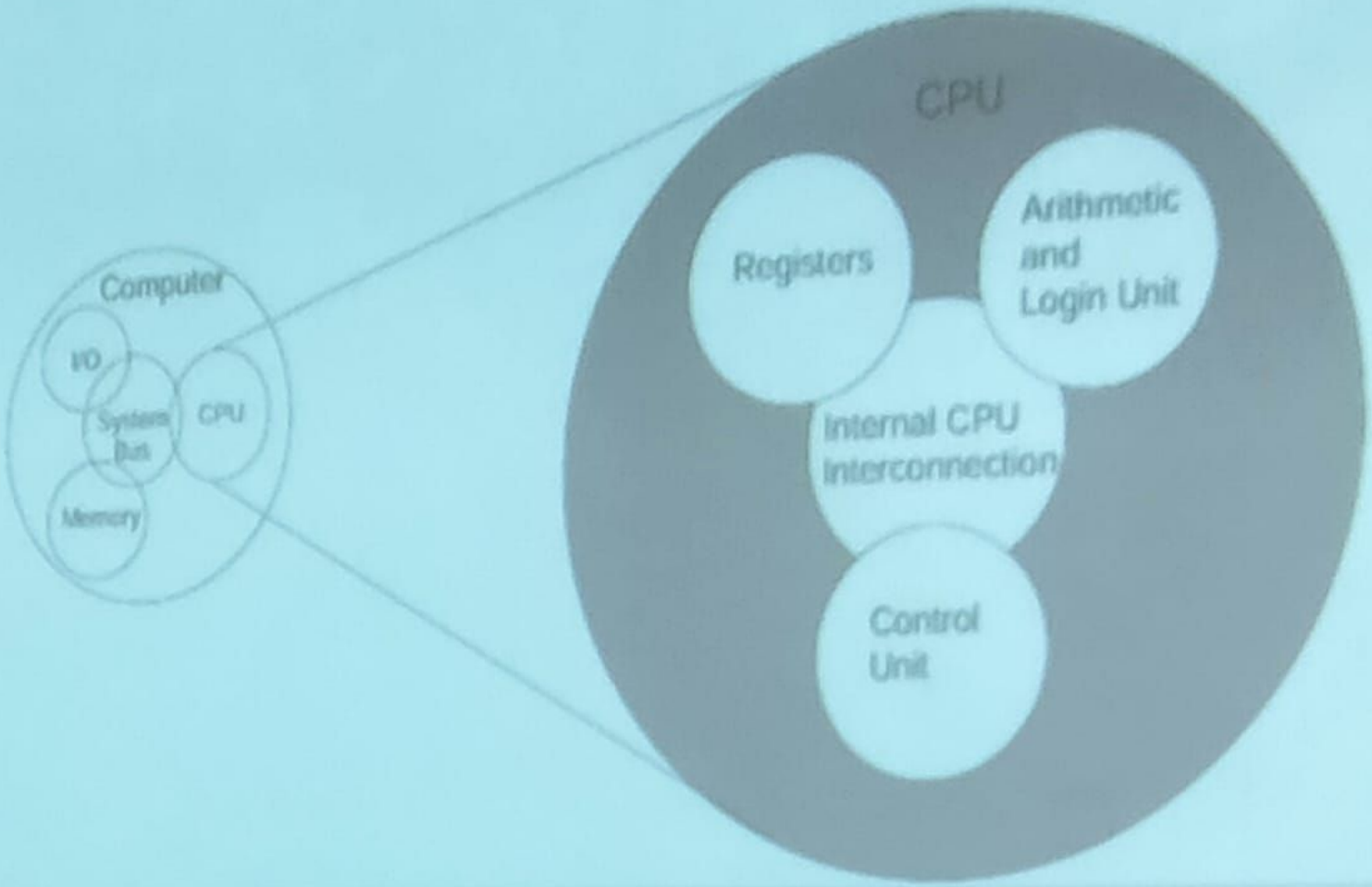
Computer Structure - Top Level



The CPU Structure



Arithmetic Logic Unit

- *The Arithmetic Logic Unit (ALU) is the heart of any CPU.*
- An ALU is a combinational digital circuit that performs both arithmetic and logic operations.
- ALUs that perform multiplication and division are designed around the circuits developed for these operations while implementing the desired algorithm.
- *In some processors, the ALU is divided into two units: an arithmetic unit (AU) and a logic unit (LU).*
- Some processors contain more than one AU -- for example, one for fixed-point operations and another for floating-point operations.
- More complex ALUs are designed for executing Floating point, Decimal operations and other complex numerical operations.
- *In computer systems, floating-point computations are sometimes done by a floating-point unit (FPU) on a separate chip*
- *These are called numeric Coprocessors*
- They work in tandem with the main processor.

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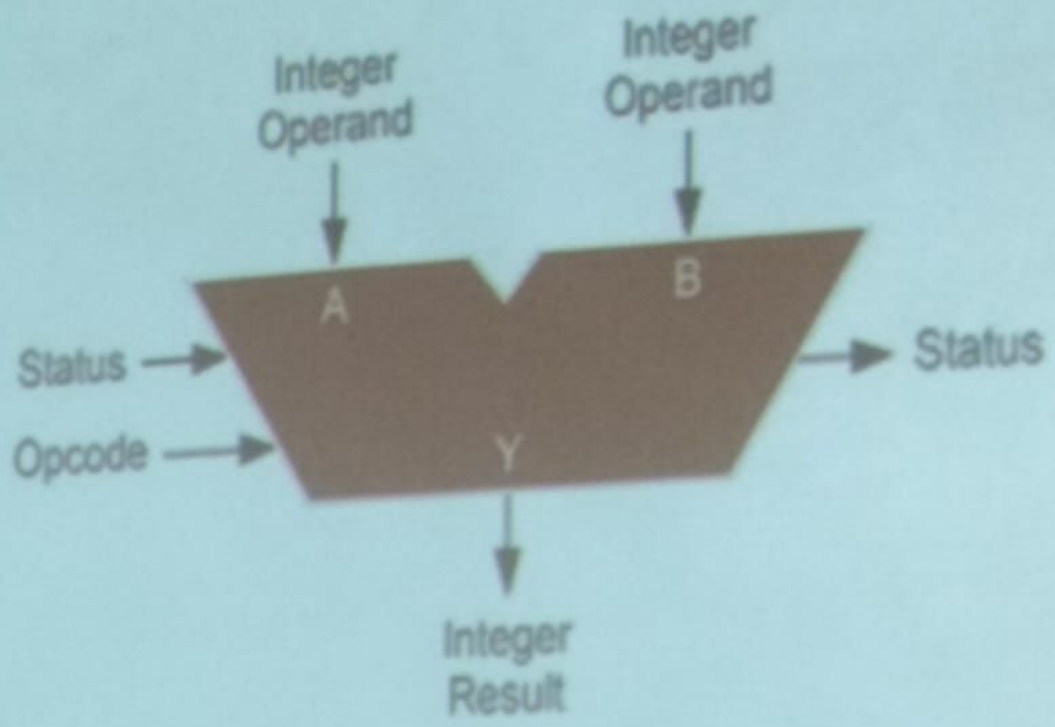
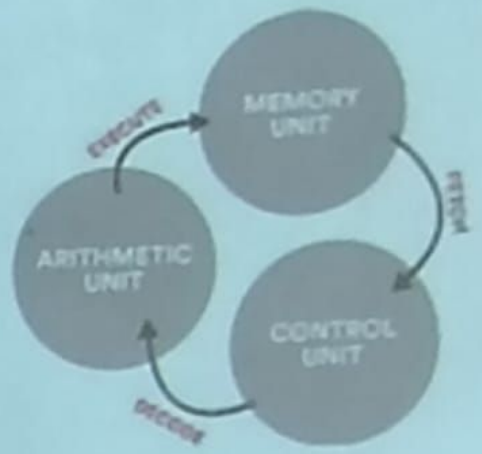
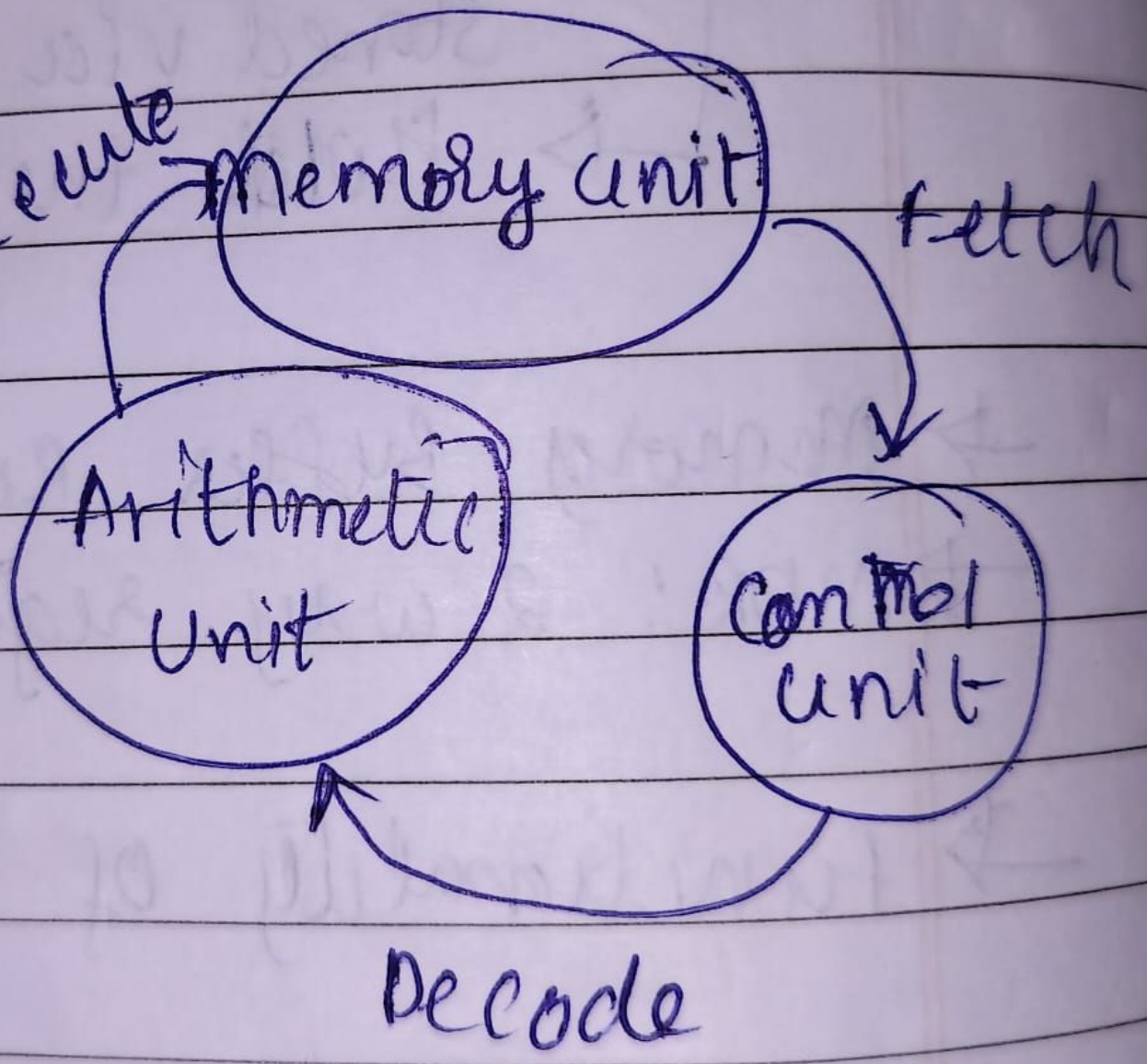
ALU Block Diagram



ALU Operations

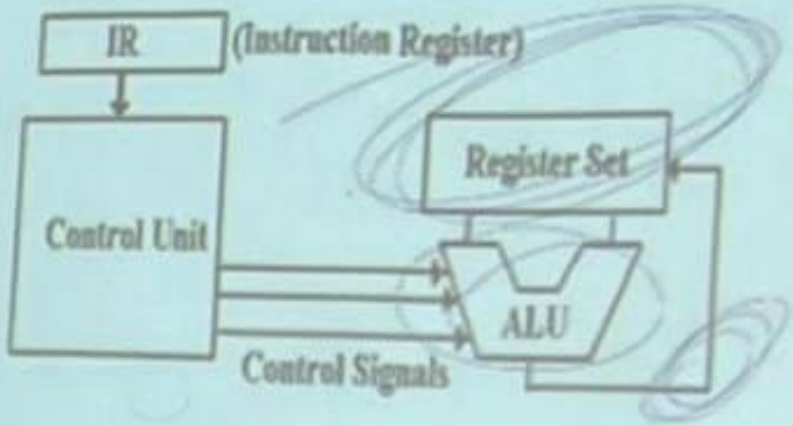
- ALU is the *"calculator"* portion of the computer.
- An ALU performs *three kinds* of operations, i.e.
  - **Arithmetic operations** such as Addition/Subtraction, Increment/Decrement, 2's Complement, etc.
  - **Logical operations** such as bitwise AND, OR, Ex-OR, 1's Complement
  - Data movement (bit shift) operations such as Load and Store



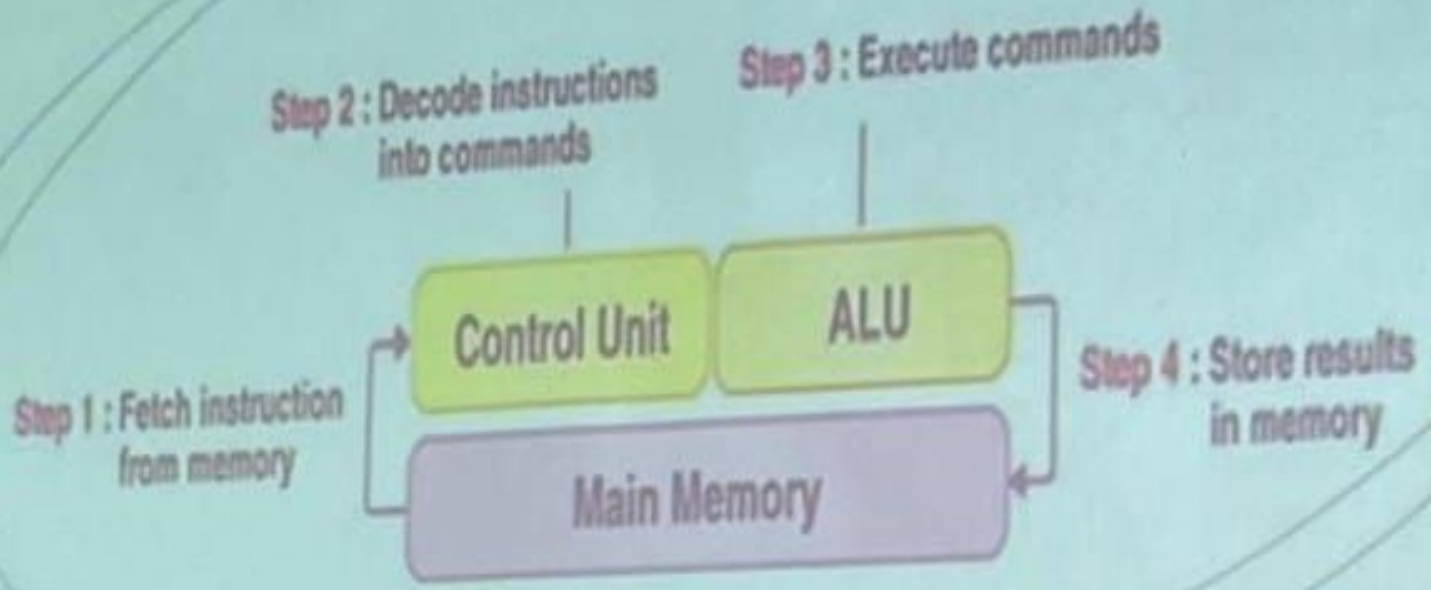

execute
memory unit
fetch
Arithmetic Unit
Control unit
Decode

ALU Workings

- Typically, the ALU has direct input and output access to the processor controller, main memory and input/output devices.
- Inputs and outputs flow along an electronic path that is called a **bus**.
- The input consists of an instruction word, sometimes called a machine instruction word.
- It contains an operation code or "opcode," one or more operands and sometimes a format code.
- The operation code tells the ALU what operation to perform and the operands are used in the operation.
- For example, two operands might be added together or compared logically.
- The format may be combined with the opcode and tells, for example, whether this is a fixed-point or a floating-point instruction.
- The output consists of a result that is placed in a storage register.



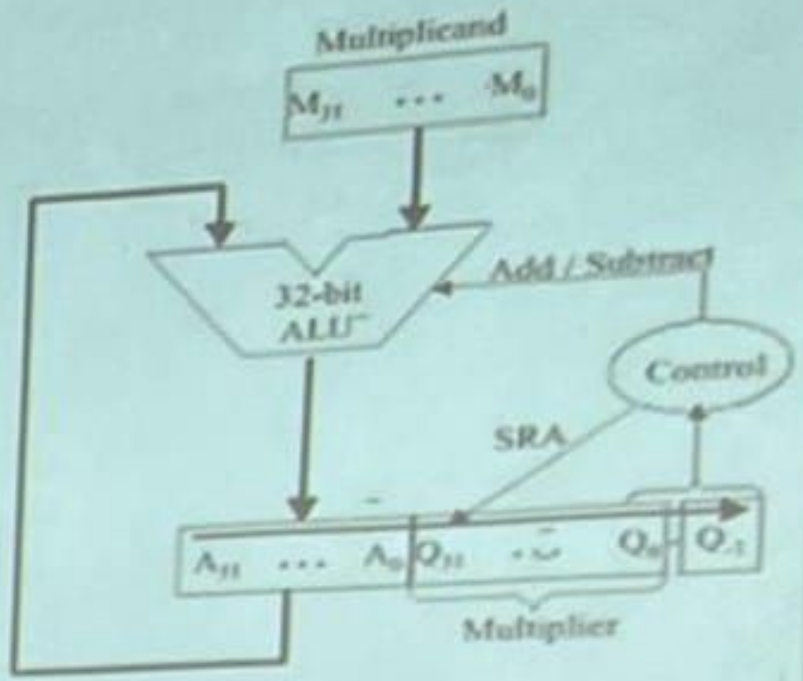
ALU Workings



Machine Cycle

Uses of ALU

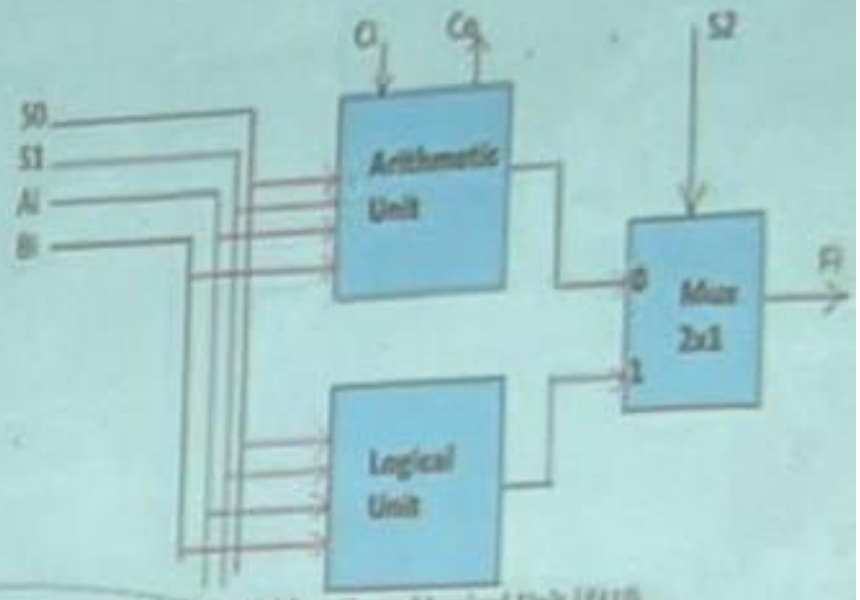
- ALUs, in addition to doing addition and subtraction calculations, also handle the process of multiplication of two integers because they are designed to perform integer calculations; thus, the result is likewise an integer.
- Division operations, on the other hand, are frequently not done by ALU since division operations can result in a floating-point value.
- Instead, division operations are normally handled by the floating-point unit (FPU), which may also execute other non-integer calculations.
- Computer Architects can also design the ALU to do any operation they choose.
  - However, as the operations become more sophisticated, ALU becomes more expensive since it generates more heat as well as takes up more space on the CPU.
  - Therefore, architects create powerful ALUs, ensuring that the CPU is both quick and powerful.



Uses of ALU

- The ALU performs the computations required by the CPU; most of the operations are logical in nature.
- If the CPU is built more powerful, it will be designed on the basis of the ALU.
  - *Then it generates more heat and consumes more energy or power.*
  - As a result, there must be a balance between how intricate and strong ALU is and how much it costs.
- The primary reason why faster CPUs are more expensive is that they consume more power and generate more heat due to their ALUs.

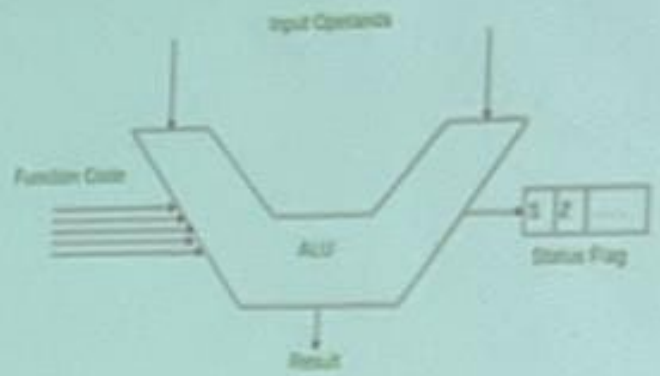
The ALU's major functions are arithmetic and logic operations, as well as bit-shifting operations.

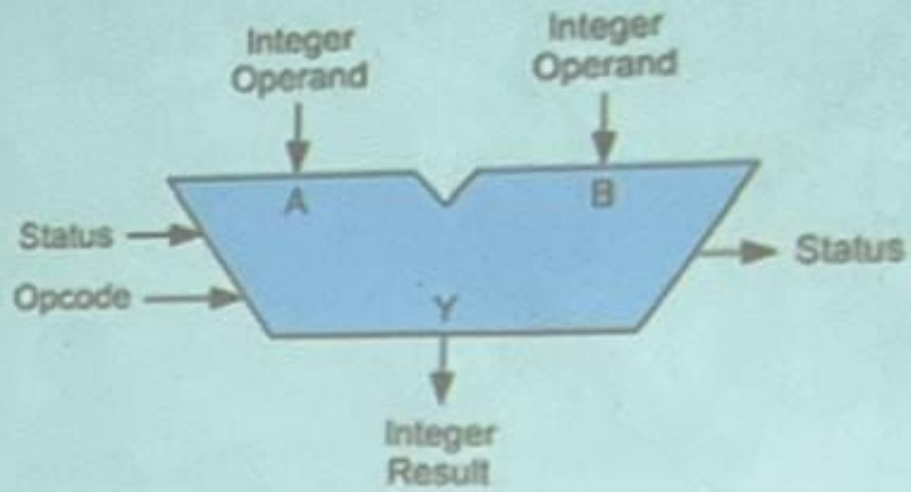


Single Stage Arithmetic and Logical Unit (ALU)

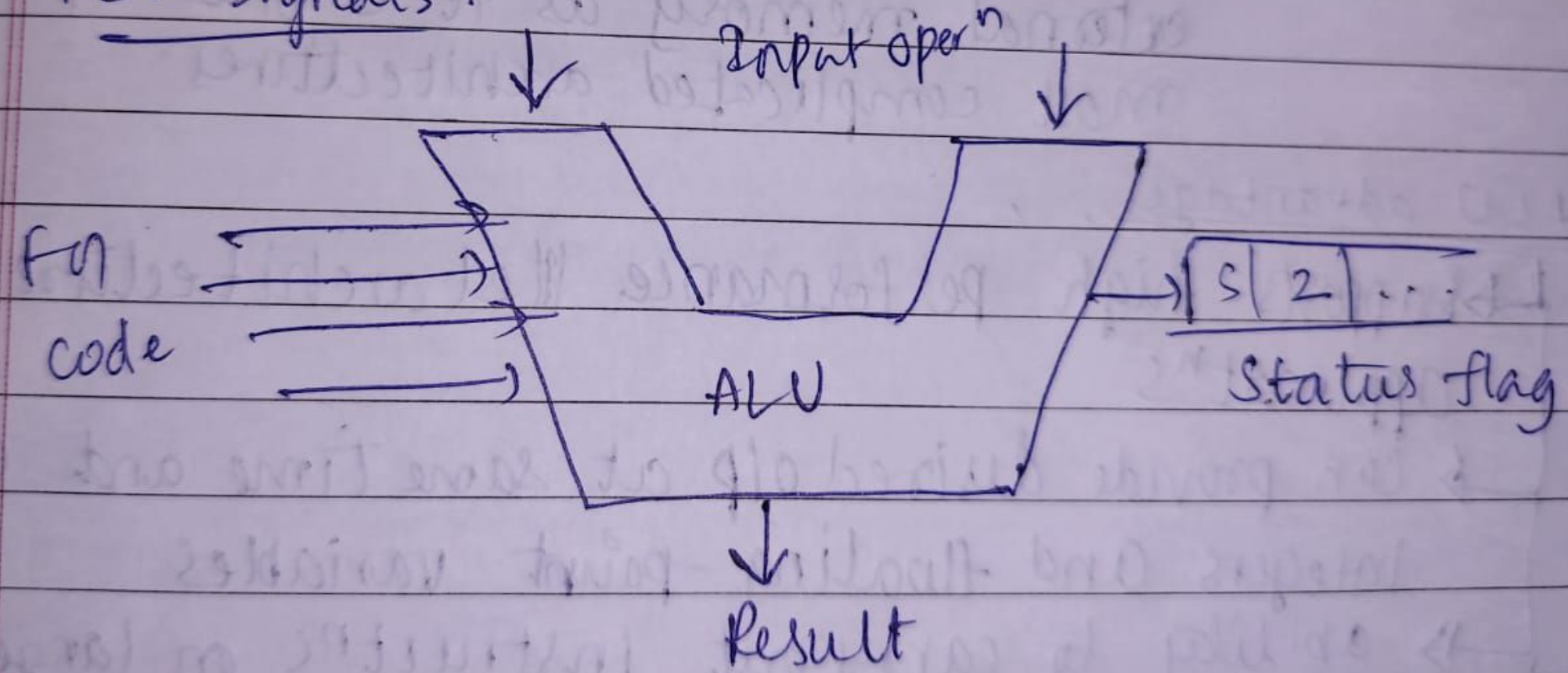
ALU Signals

- The ALU contains a variety of electrical input and output connections, which result in the digital signals being cast between the ALU and the external electronics.
- External circuits send signals to the ALU input, and the ALU sends signals to the external electronics.





→ ALU signals:



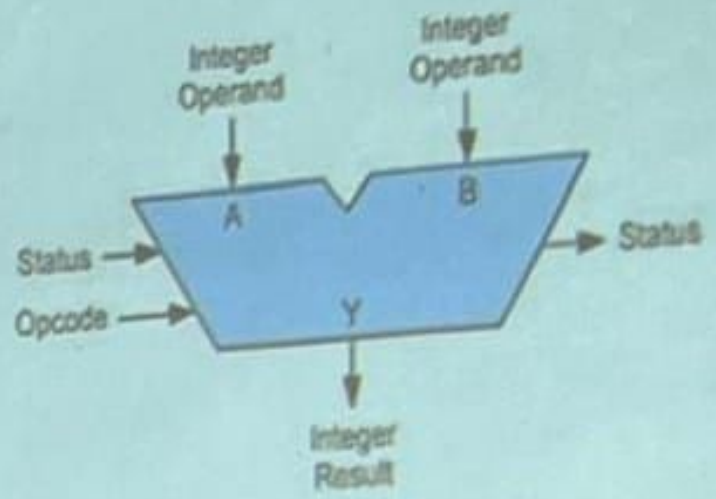
ALU Signals

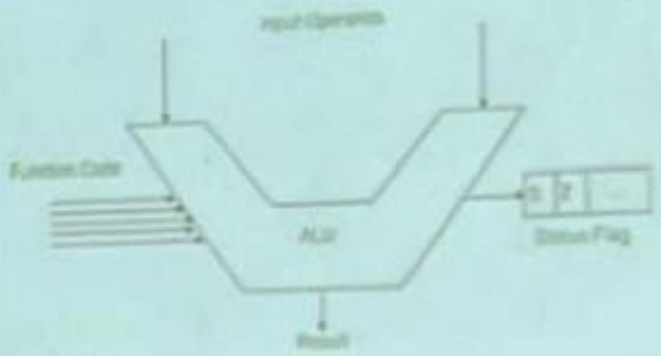
- The ALU contains a variety of electrical input and output connections, which result in the digital signals being cast between the ALU and the external electronics.
- External circuits send signals to the ALU input, and the ALU sends signals to the external electronics.

A. **Opcode:** *The operation selection code specifies whether the ALU will conduct arithmetic or a logic operation when it performs the operation.*

B. **I/O Data:** The ALU contains three parallel buses, each with two input and output operands. These three buses are in charge of the same amount of signals.

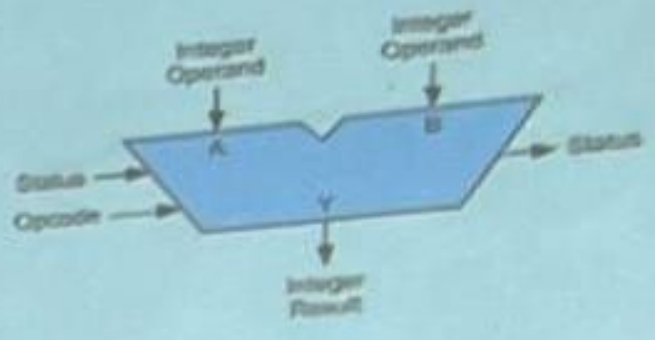
C. **Status: Input and Output**



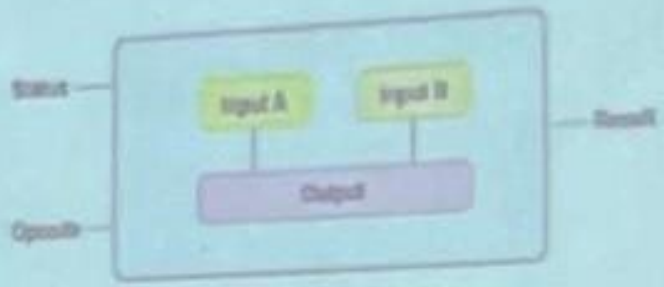


ALU Signals

- **Input Status :**
  - ➢Once ALU has completed the operation, then the status inputs allow the ALU to obtain more information needed to complete the process successfully.
  - ➢A single "carry-in" is used, which is a stored carry-out from the prior ALU operation.

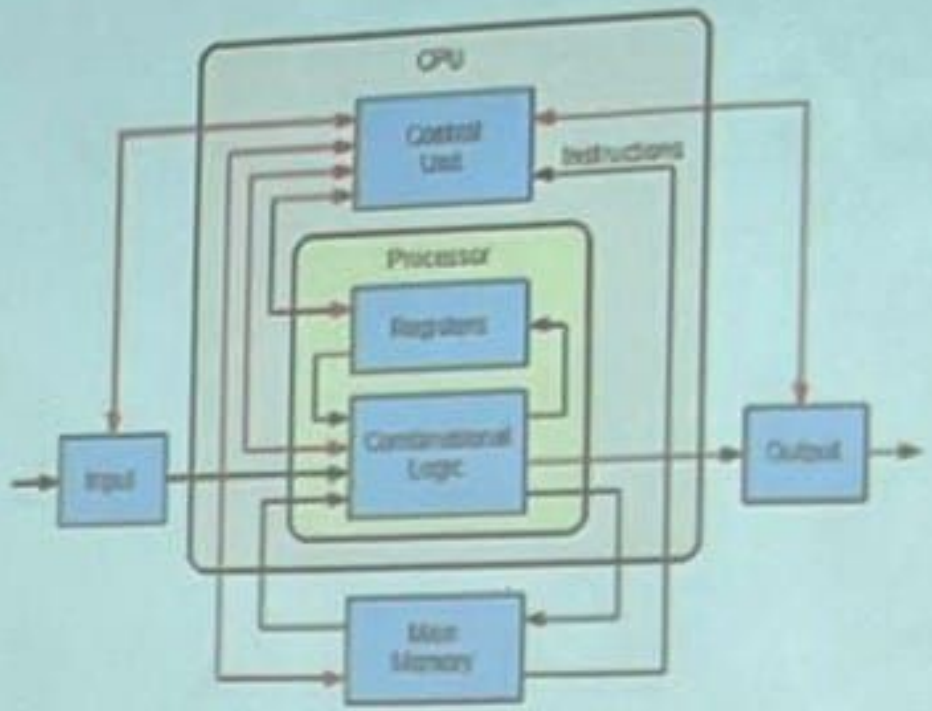


- **Output Status:**
  - ➢The status outputs, which are numerous signals, offer the results of an ALU operation in the form of extra data.
  - ➢Overflow, carry out, zero, negative, and other status signals are usually handled by general ALUs.
  - ➢The status output signals are stored in the external registers after the ALU completed each operation.
  - ➢These signals are saved in external registers, which allows them to be used in future ALU operations.



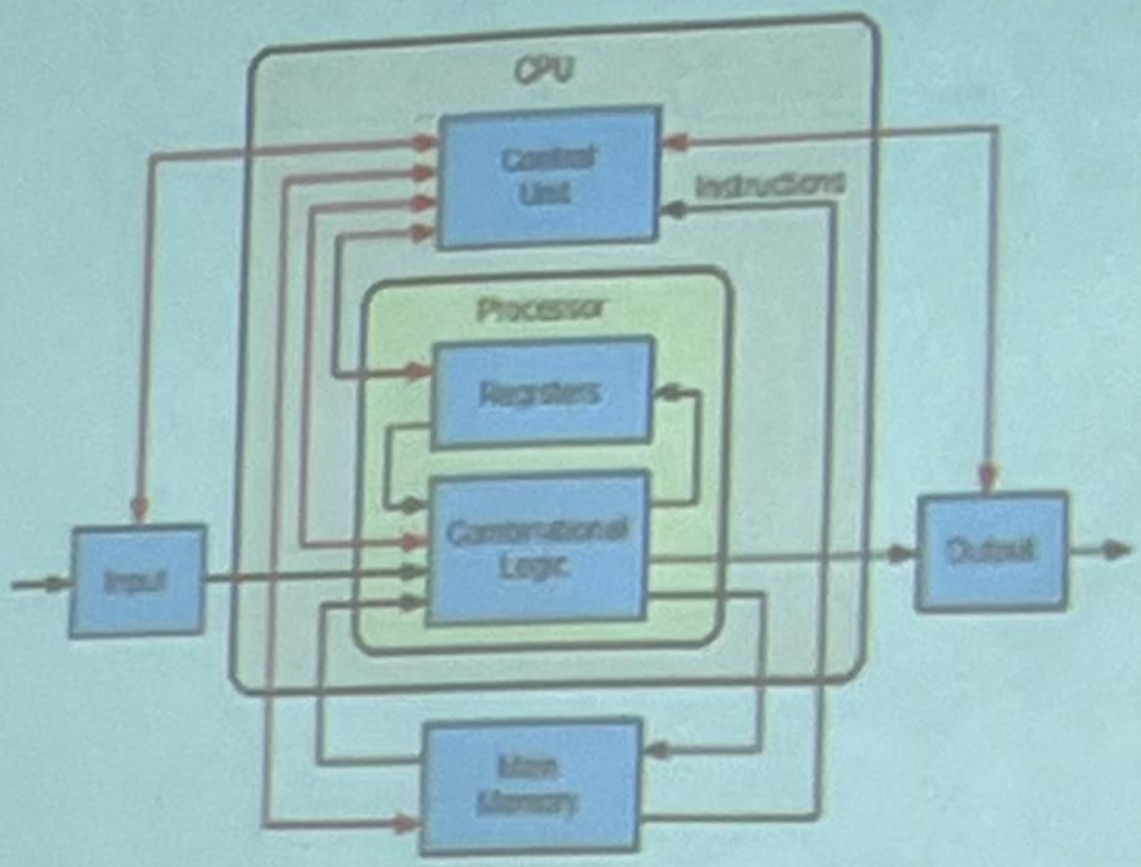
ALU Configurations

- ALU interacts with the processor.
- These configurations are included in every arithmetic logic unit:
  - Accumulator
  - Instruction Set Architecture
  - Stack Register
  - Register Stack
  - Register to Register
  - Register Memory



figurations

etic logic unit:

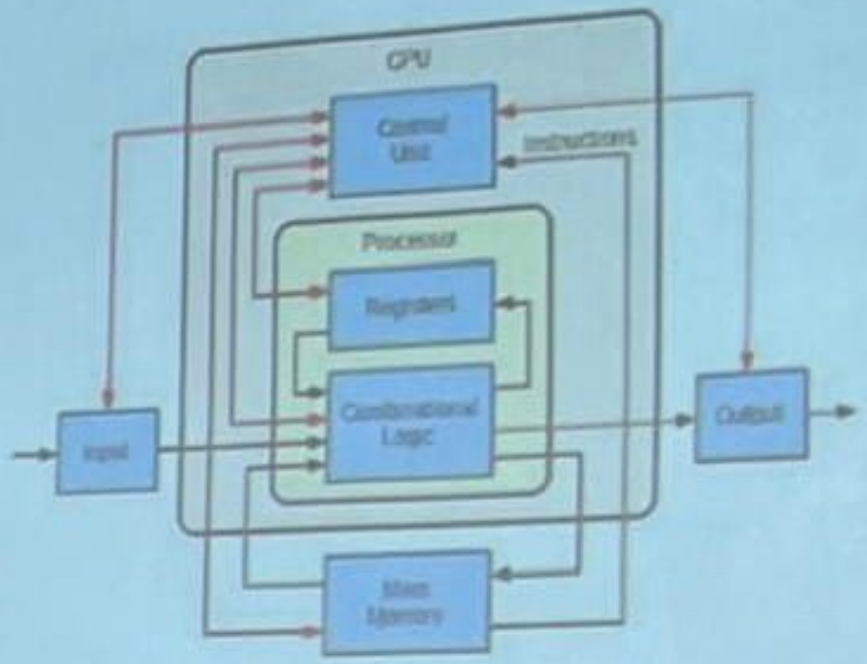


ALU Configurations

- *Accumulator*: The intermediate result of every operation is contained by the accumulator

- Instruction Set Architecture:

- *Stack Register:* the latest operations are performed, these are stored on the stack that holds programs in top-down order, which is a small register.

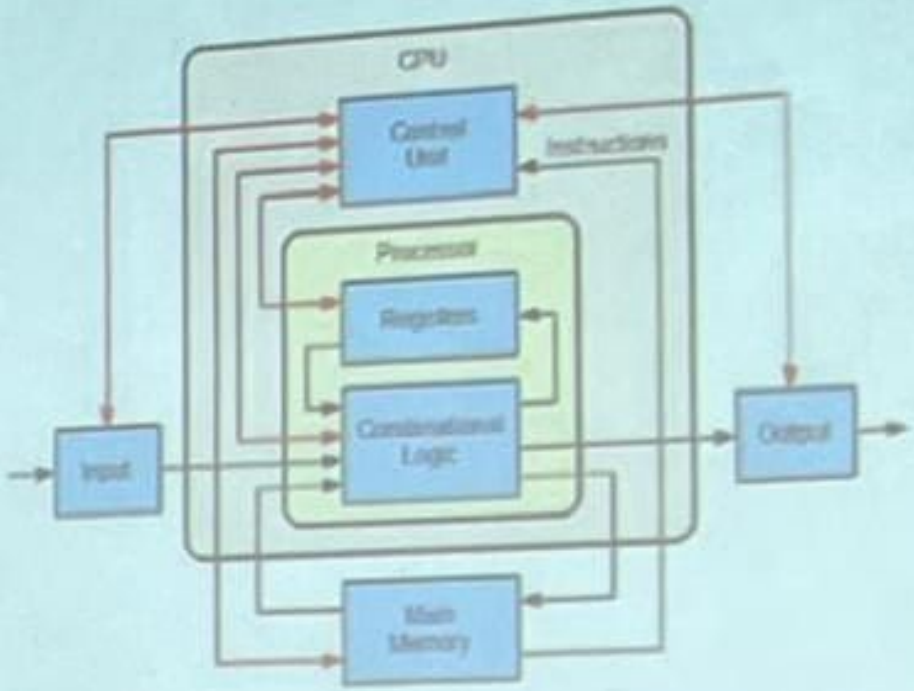


ALU Configurations

- Register - Stack:
  - Generally, the combination of Register and Accumulator operations is known as for Register - Stack Architecture.
  - The operations that need to be performed in the register-stack Architecture are pushed onto the top of the stack.

- *Register to Register:* It includes a place for 1 destination instruction and 2 source instructions, also known as a 3-register operation machine.

Register - Memory:
- In this architecture, one operand comes from the register, and
- the other comes from the external memory as it is one of the most complicated architectures.



ALU Advantages

- Multiple benefits of using ALU in CPUs

1. It supports high-performance parallel architecture and applications.
2. It can provide the desired output at the same time and combine integers and floating-point variables.
3. It has the ability to carry out instructions on a large number of items and has a high level of precision.
4. The ALU can combine two arithmetic operations in the same code, such as multiplication and addition or subtraction and addition, or any two operands.
   - E.g., $A+B*C$
5. They remain consistent throughout the presentation, and they're spaced in such a way that they do not interrupt any of the segments.
6. It is, in general, highly rapid, and as a result, it produces results swiftly.
7. With ALU, there are no sensitivity difficulties or memory wastage.
8. They are less costly and reduce the number of logic gates required.

ALU Disadvantages

- ALU's drawbacks include:
  1. Floating variables have higher delays with the ALU, and the intended controller is difficult to grasp.
  2. If memory space were fixed, bugs would appear in the results.
  3. Sometimes their circuits are complex, and the principle of pipelining is also difficult to grasp.
  4. The inconsistencies in latencies are a known drawback of ALU.
  5. Another flaw is rounding off, which reduces precision.